



Fig. 32: ACDS display (Image courtesy: US Navy)

CSG commander is issued an order to neutralise that base. This is done through GCCS-M. For such information exchanges, datalinks TADIXS and OTCIXS play a vital role.

The CSG commander can also access information through special wireless Internet systems. CSG is a node in secure internet protocol router network (SIPRNET), non-secure internet protocol router network (NIPRNET) and joint worldwide intelligence communication system (JWICS).

SIPRNET is a military Internet system owned by Department of Defence (DoD) of the USA. It has only restricted access. Through this Internet, classified information is passed through packet switching.

NIPRNET is a type of Internet that is used to exchange sensitive but non-secret information between privileged users and at the same time provides them Internet access. Again, Internet Protocol routers are owned by DoD of the USA.

JWICS is DoD's, top-secret version of the Internet, together with its secret counterpart, SIPRNET.

Previously, GCCS-M was implemented through high-performance UNIX workstations because of their capability to run GCCS-M specific software. The trend of availability of increasing computing power and increasing software versatility is leading to a migration of GCCS-M towards off-the-shelf systems.

ACDS. This is a centralised,

for the CSG.

ACDS is a comprehensive system and contains many computers, a display system with many consoles. It integrates on a mainframe computer the information from carrier's sensor systems, datalinks, navigation/air traffic control systems and weapon systems. This system is operated by 23 crew members.

Computers running application-specific software collate and correlate data. All surface and air search radar tracks and sonar tracks are communicated to the ACDS by Aegis ships of the CSG through datalinks. The respective ships derive their coordinates from their navigation systems called inertial navigation system, cross referred by a GPS.

This they send with their identification data while sending track data. ACDS, after receiving all data, correlates it with various similar tracks and positional coordinates of that ship. For example, two ships present in the CSG but sailing at different locations might have sent track data of the same enemy aircraft to the ACDS. In reality, these tracks are nothing but the track of the same aircraft taken from different locations.

ACDS, after correlation, classifies that track as the same aircraft by comparing the positional coordinates sent by the ships. This is called grid-locking. Like this, the ACDS displays the tracks accord-

ing to provide an overall tactical picture around the CSG. This gives the CSG commander the capability to identify targets, classify these and prioritize threats for easy decision making.

Based upon the threat, the commander initiates engagements like vectoring combat aircraft to targets, or launching SAM from a destroyer, or launching anti-ship missile from a destroyer.

Generally, all constituent ships of the CSGs are delegated with responsibilities to deal with threats occurring in their sectors. In one Aegis cruiser, the air defence command centre of the CSG, there is an air defence warfare commander working full-time. His or her commands are executed by the captains of Aegis cruisers and destroyers.

Similarly, there are commanders for underwater warfare, surface warfare, EW and strike warfare. The CSG commander can also exchange targeting information and engagement orders, not only within his or her CSG but also with other CSGs.

Cruise-control operations

Cruise-control operations involve navigation and propulsion operations, and are carried out in a similar way as commercial shipping navigation.

Navigation. Navigation was done through compass in the olden days; then emerged sextants and printed navigation charts. These printed navigation charts are like a detailed map but for sea routes, and indicate depth of a certain area in the sea, width of straits and other information. Now, these charts are the tools of the past and are only used as a fallback option today. Their roles have been taken over by a system called electronic chart display and information system (ECDIS), which takes inputs from electronic navigation charts (ENCs) and displays the area of interest.

These ENCs have complete